

Functions

Definition: A function f from a set P into a set Q is a relation from P to Q such that each element of P is related to exactly one element of the set Q .
It is denoted by $f: A \rightarrow B$ and read as " f is a function from P to Q ".

Note: (1) There may be some elements of the set Q which are not related to any element of set P .
(2) Every element of P must be related with at least one element of Q .

If a set A has n elements, then there are n^n functions from A to A .

Similarly, if A has m elements and B has n elements, then total number of functions from A to B are n^m and total number of functions from B to A are m^n .

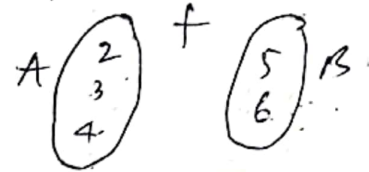
Ex (1): If $A = \{2, 3, 4\}$, and $B = \{5, 6\}$.
Determine all the functions from A to B .

(1)

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Solution: Total number of functions from A to B are $2^3 = 8$

(i) $\{(2,5), (3,5), (4,5)\}$



(ii) $\{(2,6), (3,6), (4,6)\}$

(iii) $\{(2,5), (3,5), (4,6)\}$ (iv) $\{(2,5), (3,6), (4,6)\}$

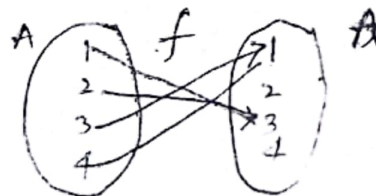
(v) $\{(2,6), (3,5), (4,5)\}$ (vi) $\{(2,6), (3,6), (4,5)\}$

(vii) $\{(2,6), (3,5), (4,6)\}$ (viii) $\{(2,5), (3,6), (4,5)\}$

Domain of a function: Let f be a function from P to Q . The set P is called domain of the function. The set Q is called Codomain of the function.

Image of an element: If the element x of P corresponds to y under function f , then y is the image of x under f and written as $f(x) = y$.

Example: Let $A = \{1, 2, 3, 4\}$ and the function $f: A \rightarrow A$ as shown in figure



(2)

(8)

(i) the image of each element of A is $\frac{3}{A}$
 $f(1)=3, f(2)=3, f(3)=1, f(4)=1$

(iii) the image of $f(A)$ of the function f :
 $f(A) = \{1, 3\}$.

Example 2: Let $P = \{x, y, z, u\}$, $Q = \{a, b, c, d\}$
and $f: P \rightarrow Q$ such that $f = \{(x, a), (y, b), (z, c), (u, c)\}$.

the domain of f is the set P .
Codomain of f is the set Q .
Range of f is $\{a, b, c\}$.

Types of functions:

(a) Injective (one-to-one) functions

Let $f: X \rightarrow Y$. The function f is called injective or one-to-one if different elements in X have different images in Y i.e.

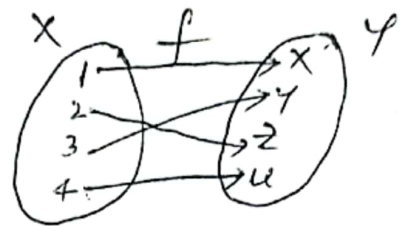
$$f(a) = f(a') \Rightarrow a = a' \quad \forall a, a' \in X$$

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or every element of domain X has a unique image in the codomain Y and there is no element of Y which is image of more than one element of domain X .

Example: Consider $f: X \rightarrow Y$.

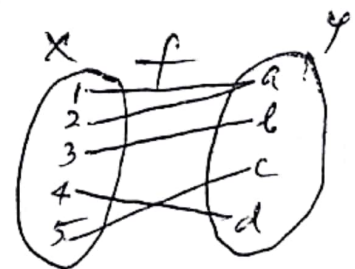


the function f is injective as every element of domain X has a unique image in the codomain.

(b) Surjective (onto) functions

Let $f: X \rightarrow Y$. the function f is called surjective if each element in Y is the image of at least one element in X .

Example: Consider $f: X \rightarrow Y$.



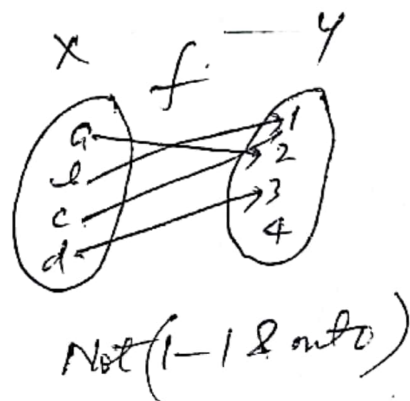
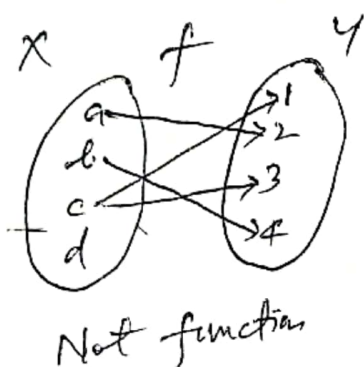
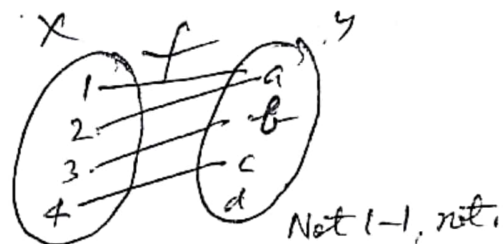
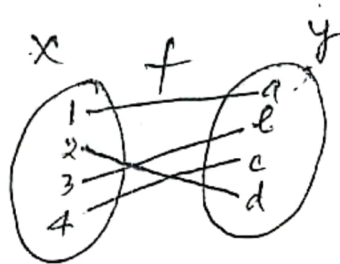
It is surjective function, as every element of Y is the image of some of X .

(c) Bijjective (one-to-one onto) functions

A function which is both (one-to-one) injective and surjective (onto) is called a bijective function.

Examples

1-1 & onto



Example: $f: \mathbb{Z} \rightarrow \mathbb{Z}$

(i) $f(n) = n - 1$

(ii) $f(n) = n^3$

(iii) $f(n) = \left\lceil \frac{n}{2} \right\rceil$

(iv) $f(n) = n$

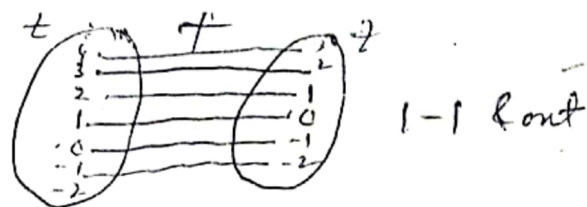
(v) $f(n) = 2n$

(vi) $f(n) = n^2$

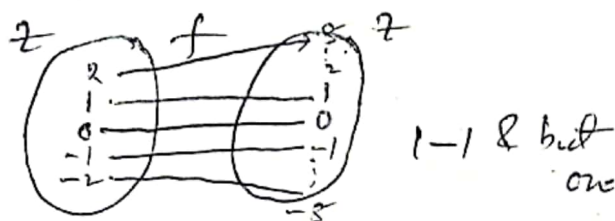
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$$f: \mathbb{Z} \rightarrow \mathbb{Z}$$

(i) $f(n) = n - 1$



(ii) $f(n) = n^3$



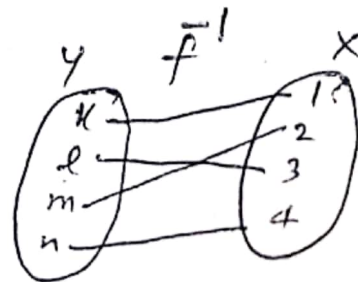
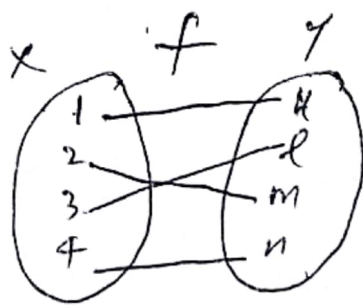
(iii) $f(n) = 2n$ 1-1 but not onto
yourself

1. The functions f and h are called equal functions if and only if $f(a) = h(a)$, $\forall a \in A$.

2. The function f is called the identity function if each element of set A has image on itself.
 i.e. $f(a) = a$, $\forall a \in A$.

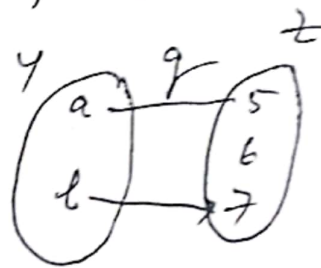
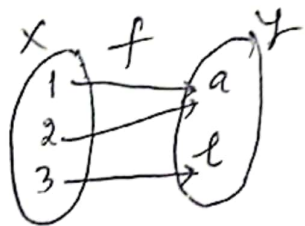
3. A function $f: X \rightarrow Y$ is invertible if and only if it is a bijective function.

Example:

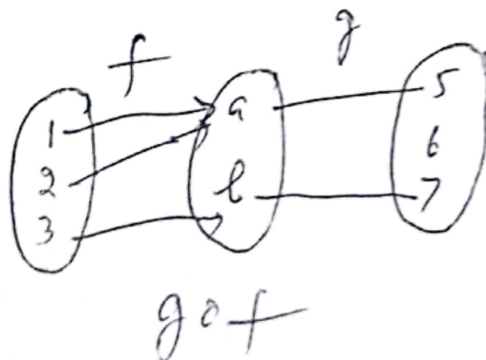


Composition of functions: Consider function $f: A \rightarrow B$ and $g: B \rightarrow C$. The composition of f with g is a function from A into C defined by $(g \circ f)x = g[f(x)]$ is denoted by $g \circ f$.

Example: Let $f: x \rightarrow y$ and $g: y \rightarrow z$ be function. Find the composition of $g \circ f$.



$$\left. \begin{aligned} (g \circ f)(1) &= g[f(1)] = g(a) = 5 \\ (g \circ f)(2) &= g[f(2)] = g(a) = 5 \\ (g \circ f)(3) &= g[f(3)] = g(b) = 7 \end{aligned} \right\}$$



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Example: Find $f \circ g$ and $g \circ f$, where
 $f(x) = \sqrt{4+x^2}$ and $g(x) = x + \frac{1}{4}$

$$(g \circ f)(x) = g[f(x)] = g(\sqrt{4+x^2}) = \sqrt{4+x^2} + \frac{1}{4}$$

$$(f \circ g)(x) = f[g(x)] = f\left(x + \frac{1}{4}\right) = \sqrt{4 + \left(x + \frac{1}{4}\right)^2}$$

Example: $f(x) = x+2$, $g(x) = \frac{1}{x+3}$

Find (i) $(f \circ g)(1)$ and $(g \circ f)(1)$

(ii) $(f \circ g)(2)$ " $(g \circ f)(2)$

(iii) $(f \circ g)(3)$ " $(g \circ f)(3)$

(iv) $(f \circ g)(4)$ " $(g \circ f)(4)$

Sol: (i) $(f \circ g)(1) = f[g(1)] = f\left(\frac{1}{4}\right) = \frac{1}{4} + 2 = \frac{9}{4}$

(ii) $(g \circ f)(1) = g[f(1)] = g(3) = \frac{1}{6}$

(iii) $(f \circ g)(2) = f[g(2)] = f\left(\frac{1}{5}\right) = \frac{1}{5} + 2 = \frac{11}{5}$

(iv) $(g \circ f)(2) = g[f(2)] = g(4) = \frac{1}{7}$

(8)

Floor & Ceiling function:

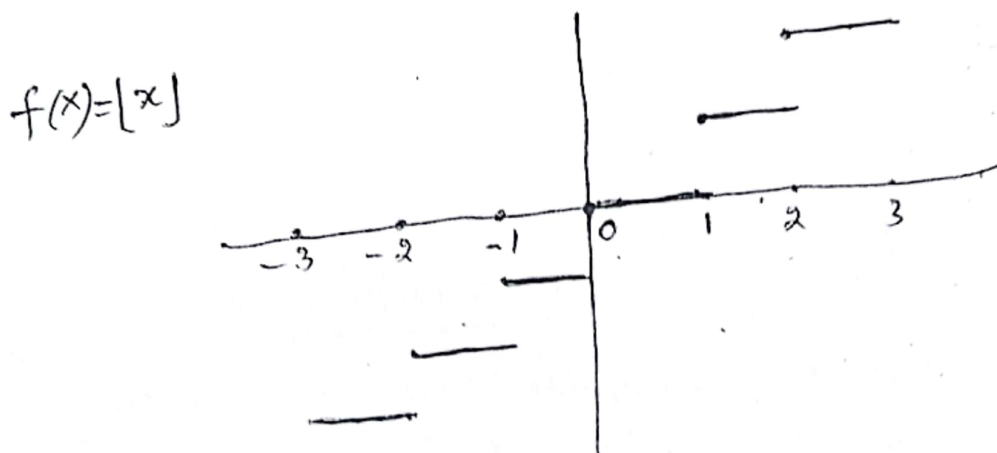
Let $x \in \mathbb{R}$. If x lies between two consecutive integers $n < x < n+1$, then n is called floor of x , denoted by $\lfloor x \rfloor$ and $n+1$ is called ceiling of x , denoted by $\lceil x \rceil$. If x is an integer then $\lceil x \rceil = \lfloor x \rfloor = x$.

Ex: (i) $x = 3.2$, then $\lfloor 3.2 \rfloor = 3$ and $\lceil 3.2 \rceil = 4$

(ii) $x = 7$, then $\lfloor 7 \rfloor = \lceil 7 \rceil = 7$.

(iii) $x = -5.2$, then $\lfloor -5.2 \rfloor = -6$ and $\lceil -5.2 \rceil = -5$

$$(iv) \left\lfloor \frac{1}{2} + \left\lfloor \frac{3}{2} + \frac{1}{4} \right\rfloor + \left\lfloor \frac{3}{2} + \frac{1}{5} \right\rfloor \right\rfloor = \left\lfloor \frac{1}{2} + 1 + 2 \right\rfloor = 4$$



Integer function

Let x be any real number. The integer value of x , written as $\text{INT}(x)$, converts x into an integer by deleting (truncating) the fractional part of the num

$$f(x) = \begin{cases} \lfloor x \rfloor & \text{if } x \geq 0 \\ \lceil x \rceil & \text{if } x < 0 \end{cases}$$

Ex (i) $\text{INT}(3.14) = \lfloor 3.14 \rfloor = 3$, $\text{INT}(\sqrt{5}) = 2$
 $\text{INT}(-8.5) = -8$, $\text{INT}(7) = 7$.

Absolute Value function

Let x be any real number. The absolute value of x , written as $\text{ABS}(x)$ or $|x|$, is defined as the greater of x or $-x$.

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Ex: $|-15| = 15$, $|7| = 7$, $|-3.33| = 3.33$, $|4.4| = 4.4$

Logarithmic functions

Let b be a positive integer. The logarithm of any positive number x to the base b , written $\log_b x$ represents the exponent to which b must be raised to obtain x . i.e. $y = \log_b x$ and $b^y = x$ (10)

are equivalent statements. Accordingly

$$\log_2 8 = 3 \text{ since } 2^3 = 8, \quad \log_{10} 100 = 2, \text{ since } 10^2 = 100$$

$$\log_2 64 = 6 \text{ since } 2^6 = 64,$$

$$\log_b 1 = 0 \text{ since } b^0 = 1$$

$$\log_b b = 1 \text{ since } b^1 = b.$$

The ~~logarithm~~ logarithm of a negative number and the logarithm of 0 are not defined.

Frequently, logarithms are expressed using approximate values. For example, using tables or calculators, one obtains $\log_{10} 300 = 2.4771$ and $\log_e 40 = 3.6889$ as approximate answers. (Here $e = 2.718281 \dots$).

Three classes of logarithms are of special importance:

Logarithms to base 10, called common logarithms;

Logarithms to base e , called natural logarithm;

and logarithms to base 2, called binary logarithm.

$$\text{i.e. } \log_2 x = \log x \quad \log_e x = \ln x$$

Exponential function

The following definitions for integer exponents (where m is a positive integer).

$$a^m = \underbrace{a \cdot a \cdot \dots \cdot a}_{m \text{ times}}, \quad a^0 = 1, \quad a^{-m} = \frac{1}{a^m}$$

$$a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

$$\text{For example, } 2^4 = 16, \quad 2^{-4} = \frac{1}{2^4} = \frac{1}{16}, \quad (125)^{2/3} = \sqrt[3]{125^2} = \sqrt[3]{15625} = 25$$

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Factorial function:

The product of the positive integers from 1 to n , inclusive is called " n factorial" is denoted by $n!$.

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-2)(n-1)n.$$

It is also convenient to define $0! = 1$.

So that the function is defined for all non-negative integers.

Thus $0! = 1$, $1! = 1$, $2! = 2 \cdot 1 = 2$, $3! = 3 \cdot 2 \cdot 1 = 6$

$$6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$$

Also observe $5! = 5 \cdot 4! = 5 \cdot 24 = 120$

$$6! = 6 \cdot 5! = 6 \cdot 5 \cdot 4! = 6 \cdot 5 \cdot 24 = 720$$

Fibonacci Sequence:

The celebrated Fibonacci sequence (usually denoted by $F_0, F_1, F_2, \dots$) is as follows:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \dots$$

i.e. $F_0 = 0$ and $F_1 = 1$ and each succeeding term is the sum of the two preceding terms. For example the next two terms of the sequence are $34 + 55 = 89$, and $55 + 89 = 144$.

Definition: (Fibonacci Sequence)

(a) If $n = 0$ or $n = 1$, then $F_n = n$

(b) If $n > 1$, then $F_n = F_{n-2} + F_{n-1}$.

Recursively defined functions

A function is said to be recursively defined if the function definition refers to itself.

Ex: Let n denote a positive integer. Suppose a function L is defined recursively as follows:

$$L(n) = \begin{cases} 0 & \text{if } n=1 \\ L(\lfloor \frac{n}{2} \rfloor) + 1 & \text{if } n > 1 \end{cases}$$

Find $L(25)$.

$$L(25) = L(12) + 1$$

$$L(12) = L(6) + 1$$

$$L(6) = L(3) + 1$$

$$L(3) = L(1) + 1$$

$$= 0 + 1 = 1$$

$$\therefore L(25) = L(12) + 1 \\ = L(6) + 1 + 1 = L(6) + 2$$

$$= L(3) + 3$$

$$= L(1) + 4 = 4 = L(25)$$

Each time n is divided by 2, the value of L is increased by 1.

$$2^L \leq n$$

$$L(n) = \lfloor \log_2 n \rfloor$$

Exercise

① Let a and b be +ve integers. A function Q is defined recursively as follows.

$$Q(a, b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) + 1 & \text{if } a \geq b \end{cases}$$

(i) Find $Q(2, 3)$ and $Q(14, 3)$, and also $Q(5861, 7)$.

Ex (2) Let $f(x): \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} x^2 - 3x & \text{if } x \geq 2 \\ x + 2 & \text{if } x < 2 \end{cases}$$

Find $f(0)$, $f(5)$, $f(-2)$.

③ Given $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$, find (a) $g \circ f$ (b) $f \circ g$
(c) $g \circ f(2)$ (d) $g \circ f(5)$
(e) $f \circ g(3)$ (f) $f \circ g(-2)$

$$\begin{aligned} 14 &= Q(14, 3) + 1 \\ &= Q(11, 3) + 2 \\ &= Q(8, 3) + 3 \\ &= Q(5, 3) + 4 \\ &= Q(2, 3) + 5 \end{aligned}$$

(4) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \left. \begin{array}{ll} 3x-1 & \text{if } x > 3 \\ x^2-2 & \text{if } -2 \leq x \leq 3 \\ 2x+3 & \text{if } x < -2 \end{array} \right\}$$

find $f(2)$, $f(4)$, $f(-1)$, $f(-3)$.